Noshitha Padma Pratyusha Juttu

San Jose, California

noshithajuttu@gmail.com | +1 4134724306 | /in/noshitha-juttu | github.com/Noshitha | hf.co/Noshitha98

PROFESSIONAL SUMMARY :

Forward Deployed AI Engineer with 3+ years building and operating client-facing enterprise data and AI systems, followed by hands-on work in voice AI, tool-calling agents, memory, evaluation, and edge deployment. Embedded with enterprise stakeholders to turn ambiguous workflows into production systems processing 1M+ records per day. Strong in Python, TypeScript, Script, APIs, agent orchestration, production debugging, and communicating reliability, latency, and deployment tradeoffs to technical and business partners.

TECHNICAL SKILLS :

Languages: Python, TypeScript, SQL, Scala, Java, C++, React, Next.js, REST APIs, webhooks
Agentic & Voice AI: Tool/function calling, LLM agents, RAG, memory, LangGraph, LangChain, MCP, Faster-Whisper, Piper TTS, multi-modal
Evaluation & Reliability: A/B testing, task-success metrics, failure analysis, hallucination, guardrails, retry/fallback, latency and cost design
Data & Cloud: Spark, PySpark, Databricks, AWS, Airflow, Snowflake, Neo4j, Docker, Kubernetes, Terraform, CI/CD, batch & streaming system

EXPERIENCE:

Deloitte USI | Data & AI Engineer Consultant *Sept 2021 – Jan 2024*

- Embedded with enterprise client operators, analysts, DBAs, application engineers, and business stakeholders to translate ambiguous requirements into data models, production workflows, validation rules, and operational SLAs.
- Built and operated Spark and Databricks pipelines processing 1M+ records per day across multi-terabyte datasets for analytics, reporting, and downstream ML use cases.
- Owned 25+ production orchestration pipelines across requirements, implementation, deployment, monitoring, incident response, root-cause analysis, and stakeholder communication.
- Migrated legacy Informatica workflows to distributed cloud pipelines, reducing batch runtime by 30% through partitioning, join optimization, and execution tuning.
- Integrated REST/SOAP sources into governed AWS workflows and implemented schema validation, retries, idempotent loading, access controls, and failure-recovery paths.
- Coordinated cross-team remediation for critical production incidents and received **two Spot Awards for delivery ownership and operational excellence**.

Adobe | *Applied AI* (Advisor: ***Prof. Andrew McCallum & Franck Dernoncourt***) *Jan 2025 – May 2025*

- Built and benchmarked an on-device multilingual machine-translation pipeline across 14 language pairs, balancing translation quality, latency, memory, and mobile deployment constraints.
- Reduced model size from 75M to 23M parameters & produced 65-70% smaller quantized ONNX exports with limited quality loss.
- Improved decoding throughput by 20% and achieved sub-100 ms translation latency in mobile deployment benchmarks.
- Built a cross-platform PyTorch to ONNX Runtime/CoreML workflow with INT8/FP16 export and evaluated quality-speed tradeoffs using BLEU, TER, chrF, METEOR, COMET, BERTScore, and BLEURT.

UMass BioNLP Lab | *AI Research Engineer* (Advisor: ***Prof. Hong Yu***) *Sep 2025 – Jan 2026*

- Built a training-free multi-agent system for clinical notes using reward-guided candidate generation, ranking, & iterative refining.
- Implemented memory reuse for high-reward reasoning for retrieval failures, inconsistent reasoning, and prompt-level regressions.
- Produced structured, auditable outputs for a high-stakes domain while analyzing accuracy, reliability, and inference-cost tradeoffs.

PROJECTS:

SnapKnow

- Prototyped in a 36-hour hackathon, a voice-driven memory assistant that supports object recall and person recognition on a Samsung Galaxy S25 Ultra.
- Integrated a local speech loop using Whisper STT, Piper TTS with on-device vision and persistent SQLite memory, prioritizing privacy and offline operation.
- Benchmarked quantized edge models from 0.22 MB at approximately 2.9 ms median inference to 0.58 MB at 12–13 ms.

Automated AGENTS.md Generator

- Built an automated pipeline that statically analyzes a target repository, generates repo-specific `AGENTS.md` instructions, and evaluates whether those instructions improve coding-agent behavior.
- Improved multi-turn repo-aware execution quality: generated instructions increased task success score from 0.67 to 0.97, raised plan adherence from 0.36 to 0.76, eliminated hallucinated commands, and triggered required client regeneration in 18/20 runs.

Graph-Based Multi-Agent Clinical Reasoning System

- Built a graph-native clinical reasoning prototype that stores synthetic CKD/diabetes patient context in Neo4j and retrieves patient-specific graph bundles for LLM-based reasoning.
- Built an n8n-style webhook-triggered agent workflow that retrieves Neo4j patient context, applies specialist LLM reasoning, and synthesizes structured recommendations from graph-derived evidence.

RESEARCH & PUBLICATIONS:

RAG Is Not Enough: Why Agentic AI Needs Memory — Article, 2026
When Consensus Becomes Compliance: Measuring Sycophancy in Multi-Agent Language Model Interactions — *Under Review, ACL'26*
Text to Trust: Evaluating Fine-Tuning and LoRA Trade-offs in Language Models for Unfair Terms of Service Detection — *Oct 2025*
Development of an AI-Based Chatbot Using Deep Neural Networks — *Oct 2021*

EDUCATION:

University of Massachusetts, Amherst — **Massachusetts, USA**
M.S. in Computer Science - GPA: 3.9/4.0 — *Jan 2024 - Dec 2025*
VNR Vignana Jyothi Institute of Engineering & Technology — **Hyderabad, India**
B.Tech. in Information Technology - GPA: 9.1/10.0 — *July 2017 - July 2021*